Project Proposal

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PayPal AI Hackathon

Build What's Next with PayPal & AI

2026

October 1 - November 12

PayFlow AI

An Enterprise-Grade Platform

B2B SaaS & APIaaS

AI-Driven Financial Operations

and Real-Time Risk Management



The Problem: Poor Cash Flow Visibility



50M+ freelancers & SMBs rely on PayPal,

but their financial tooling is:

- Reactive → Records what happened;
- Manual → Hours of categorization;
- Siloed → No unified intelligence;
- Delayed → Fraud found weeks later.

Result: $250 – $500/month in lost time.



Solution: PayFlow AI

Not just a “Thinking Ledger”. An intelligence layer on top of your payment data.





Understands

AI categorizes transactions

Predicts

Forecasts 30-day cash flow

Detects

Real-time risk scoring

Advises

Generative AI summaries

Architecture Overview

A modular monolith is an architectural approach that combines the simplicity of a monolithic architecture with the benefits of modular design.



Application Layer 2:

AI Engine

Risk Engine

Forecasting



Data Layer 4:

Elasticsearch

Presentation Layer 1:

Dashboard

REST API

Webhook Listener



Integration Layer 3:

PayPal Client

ES Manager

HF Client



Pillar 1: AI Automation

Zero-shot NLP

Mistral-7B summaries

Dispute drafts

Pillar 2: Risk Management

Cryptographic webhooks

Weighted risk scoring

Automated alerting

The Four Pillars

Each pillar is independently valuable - but together, they create something greater than the sum of its parts.



Pillar 3: Predictive Analytics

Prophet forecasting

Isolation Forest

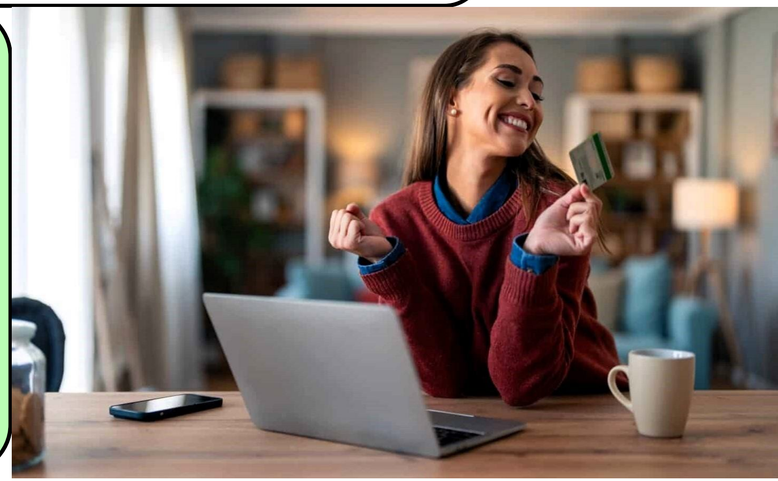
Fuzzy search

Pillar 4: Production Engineering

Full observability

155 automated tests

CI/CD pipeline



The Core Innovation: Weighted Risk Scoring

The score maps to 4 deterministic action levels

The formula fuses three heterogeneous signals into a single interpretable score from 0 to 100.





Score → Action:

-0–39 → Low → Auto Approve;

-40–59 → Medium → Monitor;

-60–79 → High → Manual Review;

-80–100 → Critical → Hold & Investigate

$R = 100 \times (0.50 \times f_1 + 0.30 \times f_2 + 0.20 \times f_3)$

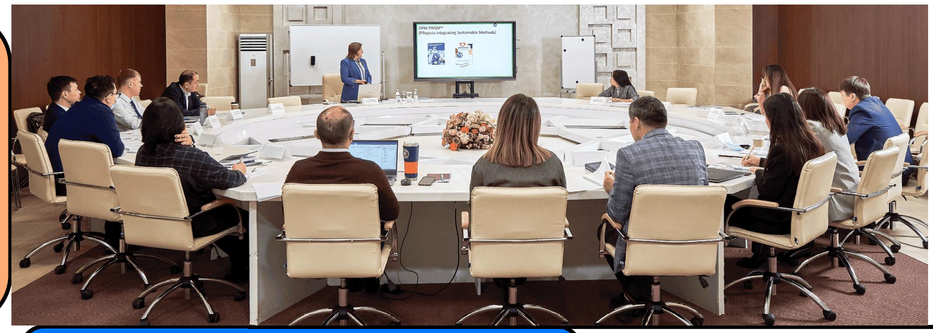
- f_1 = Anomaly score (50%)
- f_2 = 1 – AI confidence (30%)
- f_3 = Amount / (5 × avg amount) (20%)

Ingress:

RSA signature verification

Certificate domain pinning

5-minute replay protection



Authentication:

OAuth 2.0 with intelligent caching

Thread-safe token refresh

Runtime:

Rate limiting (SlowAPI)

Input validation (Pydantic v2)

Secrets masking in logs

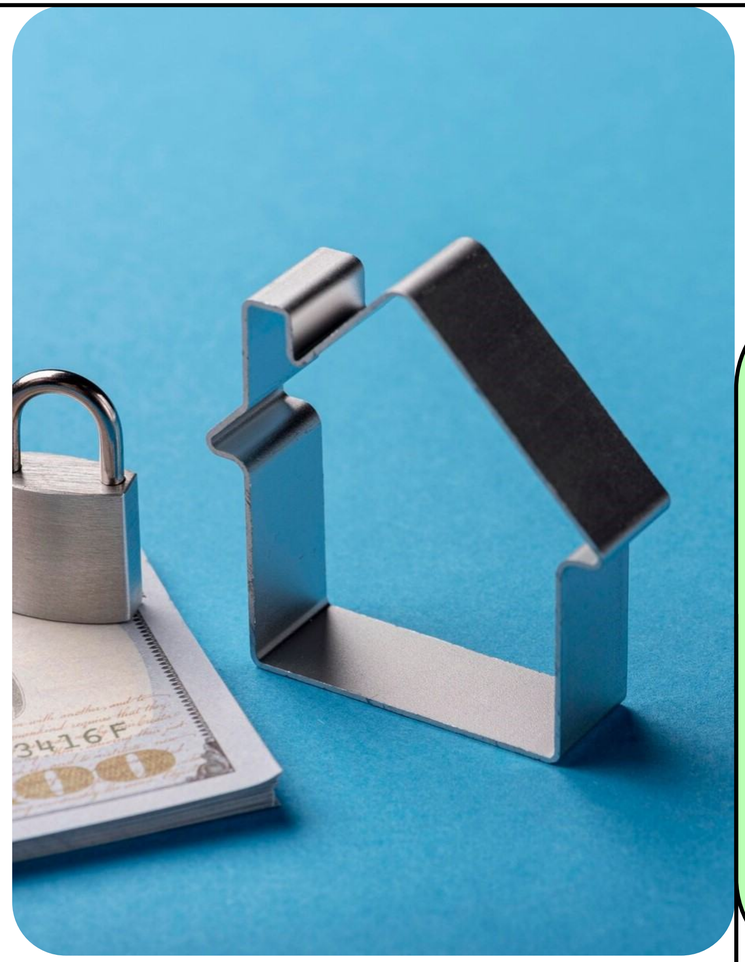
Observability:

Prometheus metrics

OpenTelemetry traces

Structured JSON logs

Security Model: Defence in Depth



Performance Benchmarks

Operation	Scale	Latency
AI categorization	1 tx	< 10 ms
Batch categorization	100 tx	< 200 ms
Batch categorization	1000 tx	< 2 s
Risk scoring	100 tx	< 500 ms
Elasticsearch search	any	< 500 ms
Prophet forecast	30 days	< 3 s
Anomaly detection	100 tx	< 2 s



Target workload: 10^2 – 10^4 transactions/day

Test coverage: 92% (155 automated tests)

Economic Impact (ROI)

Time Savings:

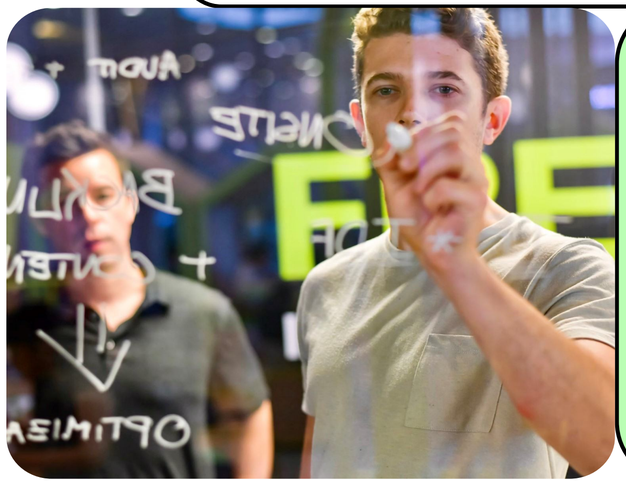
→ 5–10 hours/month of manual bookkeeping eliminated;

→ $3,000–6,000/year saved;

Cash Flow Optimization:

→ Accurate forecasting prevents overdrafts;

→ $500–2,000/year in avoided fees;



Fraud Mitigation:

→ Catching one $5K fraudulent transaction;

→ Pays for the platform for years;

Dispute Recovery:

→ AI-assisted responses increase win rates;

→ Direct revenue recovery;

Total estimated ROI: $4,000–$10,000+ per user/year

Competitive Differentiation

Where does PayFlow AI fit in the market? Existing solutions fall into two categories, and neither fully addresses the problem.



Enterprise Saas: Ramp, Brex, Expensify

✓ Polished
✗ Closed-source
✗ Expensive
✗ Corporate focus

Open-Source Personal: Firefly III, Actual Budget

✓ Free
✗ No real-time AI
✗ No risk scoring
✗ No webhook integration

PayFlow AI:

✓ Open-source & self-hostable
✓ Real-time webhook risk scoring
✓ Unsupervised anomaly detection
✓ Generative AI for summaries & disputes
✓ Fully observable architecture

Thank You for Your Attention



PayFlow AI is a proactive intelligence layer for payment data. It's built on principled, tunable, explainable AI. It's secured with defense-in-depth cryptography. It's observable at every layer. It's tested, documented, and production-considered. And it's open-source and ready for deployment. It's a thinking ledger.

